

Richitha Rekula

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Summary

Master's student in Data Science, Analytics and Engineering at Arizona State University, focused on Software Engineering, Applied AI, and AI/ML Engineering. Built end-to-end ML and generative AI systems, from data pipelines to deployed applications. Selected from 200+ applicants for a Zoom-sponsored fellowship developing Zoom Lens.

Education

Arizona State University

Master of Science in Data Science, Analytics and Engineering | GPA: 3.78/4.0

Tempe, AZ

Expected May 2027

Anurag University

Bachelor of Technology in Artificial Intelligence and Machine Learning | GPA: 3.53/4.0

Hyderabad, India

Aug 2021 – May 2025

Technical Skills

Languages: Python, SQL, Java, JavaScript, C++, C, R, MATLAB

AI/ML: TensorFlow, Scikit-learn, XGBoost, SHAP, NLTK, VADER, OpenCV

Data Science & Analytics: Pandas, NumPy, Matplotlib, Jupyter, PySpark, Power BI, Tableau

Frameworks & Libraries: Flask, Node.js, Express.js, Streamlit, PyMuPDF

Data Platforms: Databricks, Snowflake, PostgreSQL, MongoDB, SQLite

APIs & SDKs: Google Gemini API, Zoom Video SDK, Zoom Realtime Media Streams API, REST APIs

Developer Tools: Docker, Git, GitHub Actions, ServiceNow

Experience

Zoom Fellow

ASU Next Lab (in partnership with Zoom)

Tempe, AZ

Aug 2026 – Present

- Engineered **Zoom Lens**, an in-meeting AI assistant using **Zoom APIs/SDKs** and **Generative AI** to deliver private, real-time context and explanations of shared-screen content.
- Built participant-specific request pipelines in **Python**, routing responses only to requesting users without interrupting the presenter.

Deskside Support Intern

ASU Media and Immersive eXperience (MIX) Center

Mesa, AZ

May 2026 – Aug 2026

- Provisioned and maintained **Windows, macOS, and Linux** endpoints using **Jamf Pro**, hardware diagnostics, and enterprise software deployment tools.
- Managed incidents in **ServiceNow** and documented recurring issues and resolutions to improve troubleshooting across IT teams.

Research Assistant

Anurag University

Hyderabad, India

Aug 2023 – May 2025

- Audited datasets across **5+ departments** using **Excel (Power Query, VLOOKUP, Pivot Tables)**, correcting record inconsistencies to improve reporting accuracy.
- Reorganized shared files across **SharePoint** and **Google Workspace**, standardizing record structure and reducing information retrieval time by **20%**.

Projects

Support Ticket Triage & Resolution-Time Predictor | Python, XGBoost, PySpark, SHAP, Flask, Docker Jul 2026

- Built an ML pipeline on **20,000 synthetic** support tickets comparing Logistic Regression, Random Forest, and **XGBoost**; XGBoost reached **macro-F1 0.705** on priority and reduced resolution-time error to **12.2h** from a 24.4h baseline.
- Engineered text features with NLTK and VADER sentiment, added **SHAP** per-prediction explanations, and served the model through a Flask REST API (**15–40ms** responses) packaged with Docker.

Reddit Data ETL Pipeline | PostgreSQL, SQL, Docker, GitHub Actions

Feb 2026

- Built a **5-stage ETL pipeline** (extract, validate, stage, merge, check) loading **18.9M** Reddit records into PostgreSQL; COPY bulk staging and drop-and-rebuild constraints cut full-run time from 584s to **345s**.
- Enforced referential integrity at merge time, matching expected results exactly on a planted-defect test across all 18.9M records; wrote **49 automated tests** run on every push via GitHub Actions.

Research Paper Summarizer | Python, Streamlit, PyMuPDF, Google Gemini API

Feb 2025

- Built a Streamlit app that ingests papers via PDF, pasted text, or **DOI/arXiv lookup**, extracting text, figures, tables, and references with PyMuPDF and key terms with TF-IDF, without LLM-based parsing.
- Integrated the **Gemini API** for five summary styles and context-aware chat, with model fallback and streamed responses; generated APA/MLA/BibTeX citations without an LLM.

StreamBox: Web-Based OTT Platform | Node.js, Express.js, MongoDB, Docker

Dec 2024

- Built a full-stack streaming platform with a **24-endpoint** Express REST API, JWT authentication, and 5 MongoDB collections, using atomic updates and **optimistic locking** for concurrent writes.
- Verified concurrency safety with 24 automated race-condition tests and a **100-user load test** (51,633 requests, **0** errors, **0** duplicates, **0** incorrect totals).